

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1847

Blythe 230 kV
May 16, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1847. The Interconnection Customer proposes a 194.8 MW battery energy storage facility interconnecting at the Blythe 230 kV substation in the Illinois area. The request was received on August 04, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 489 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,524,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Lansing-Dycesville 100.0 kV
Contingency: Castle Afb-Sand Rdge 100.0 kV line
Loading: 148.7% (Rating: 133 MVA)
DFAX: 5.8%
Pre-project loading: 140.2%
2. Facility: Arch Smith-Matteson 100.0 kV Line
Contingency: Blythe-S Boston 230.0 kV
Loading: 125.3% (Rating: 170 MVA)
DFAX: 6.3%
Pre-project loading: 118.1%
3. Facility: Hoosier Fairview-Blythe 100.0 kV Line 1
Contingency: Hoosier Fairview-Blythe 100.0 kV Circuit 2
Loading: 115.9% (Rating: 217 MVA)
DFAX: 7.2%
Pre-project loading: 109.4%
4. Facility: Hoosier Fairview-Blythe 100.0 kV Circuit 2
Contingency: Blythe-Hoosier Fairview 100.0 kV line
Loading: 115.9% (Rating: 217 MVA)
DFAX: 7.2%
Pre-project loading: 109.4%
5. Facility: Hoosier Fairview-Blythe 100.0 kV Line
Contingency: Hoosier Fairview-Blythe 100.0 kV Circuit 1
Loading: 115.9% (Rating: 217 MVA)
DFAX: 7.2%
Pre-project loading: 109.4%

6. Facility: Spearfish Hydro-Arch Smith 100.0 kV Line
 Contingency: S Boston-Blythe 230.0 kV
 Loading: 109.9% (Rating: 172 MVA)
 DFAX: 5.2%
 Pre-project loading: 104.0%
7. Facility: Arch Smith-Spearfish Hydro 100.0 kV
 Contingency: S Boston-Blythe 230.0 kV
 Loading: 109.9% (Rating: 172 MVA)
 DFAX: 5.2%
 Pre-project loading: 104.0%
8. Facility: South Boston-Blythe 230.0 kV
 Contingency: Blythe-Lincoln 230.0 kV
 Loading: 102.9% (Rating: 714 MVA)
 DFAX: 68.1%
 Pre-project loading: 84.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lansing-Dycesville 100.0 kV	Rebuild line with higher-capacity conductor	\$1,156,000	11% (\$128,000)
Matteson-Arch Smith 100.0 kV	Rebuild line with higher-capacity conductor	\$1,871,000	11% (\$204,000)
Hoosier Fairview-Blythe 100.0 kV Circuit 1	Rebuild line with higher-capacity conductor	\$2,649,000	12% (\$308,000)
Hoosier Fairview-Blythe 100.0 kV Circuit 2	Rebuild line with higher-capacity conductor	\$3,792,000	12% (\$441,000)
Blythe-Hoosier Fairview 100.0 kV	Rebuild line with higher-capacity conductor	\$2,880,000	12% (\$335,000)
Arch Smith-Spearfish Hydro Substation 100.0 kV Circuit 1	Reconductor line	\$862,000	11% (\$94,000)
Arch Smith-Spearfish Hydro 100.0 kV Circuit 2	Reconductor line	\$1,795,000	11% (\$196,000)
South Boston-Blythe 230.0 kV	Reconductor line	\$8,696,000	21% (\$1,818,000)
	Total allocated to project		\$3,524,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.